

LawnGnomeDX

User Manual

Radio Text and File Transfer

Simple operator guide

- Send and receive text through a radio/audio interface
 - Address messages by callsign or CQ
 - Transfer files and images
- Use selectable RX and TX sound devices

Important: LawnGnomeDX produces and receives modem audio. Make sure your radio and audio interface are configured so the selected TX audio is actually transmitted and the radio audio reaches the selected RX input.

1. What LawnGnomeDX Does

LawnGnomeDX is a sound-card radio modem application for exchanging text and files. The main window is split into a transmit side and a receive side, with a waterfall display along the bottom.

- Send ordinary typed text messages.
- Address transmissions to another callsign or use CQ for a general call.
- Receive and display messages with sender, time, and signal-to-noise information.
- Import and transfer files and images.
- Select separate sound devices for radio receive audio and transmit audio.

2. Before You Start

1. Connect your radio or radio interface to the computer so radio receive audio reaches the computer and computer transmit audio reaches the radio.
2. Launch LawnGnomeDX. If prompted, choose the interface language you want to use.
3. Open Sound > Sound Devices and choose INPUT (RX) and OUTPUT (TX). Use Save if you want the selections remembered for the next launch.
4. Enter your station identifier in TX CALLSIGN. A transmit callsign must contain 3 to 6 letters and/or digits.
5. Set RX CALLSIGN to the station you want to call, or set it to CQ for a general call.

Addressing: TX CALLSIGN is your station identifier. RX CALLSIGN is the destination used when you transmit. For a direct message to another station, that station should have its own callsign entered in its TX CALLSIGN field. For general CQ traffic, receiving stations should use CQ in RX CALLSIGN.

3. Quick Start — Send a Text Message

6. Enter your callsign in TX CALLSIGN.
7. Enter the destination callsign in RX CALLSIGN, or enter CQ.
8. Type your message in the left-hand text box.
9. Press SEND MESSAGE.
10. Watch the transmit indicator and chunk counter. If the channel or cooldown is active, the button can show TX QUEUED until LawnGnomeDX can transmit.
11. Use STOP (TX) if you need to cancel transmission or clear queued work.

Tip: When SEND MESSAGE is showing TX QUEUED, right-clicking it cancels the most recently queued message.

4. Receive a Message

12. Make sure the correct INPUT (RX) sound device is selected and radio audio is reaching the computer.
13. For messages sent directly to you, keep your own station callsign in TX CALLSIGN. For CQ reception, set RX CALLSIGN to CQ.
14. Leave LawnGnomeDX running. The RECEIVING MESSAGE and DECODING indicators show receive activity.
15. Decoded messages appear in the right-hand pane. The display can include the sender, time, and SNR reading.

RX Pane Buttons

Control	What it does
STOP (RX)	Finish/reset the current receive session. Use it if a receive attempt is stuck or you want to abandon the current decode.
COPY TEXT	Copy the latest decoded message to the system clipboard.
SAVE TEXT	Save the contents of the RX pane to a text or log file.
ERASE TEXT	Clear the RX pane, including unsaved received-file links and image previews.

5. Main Window Controls

Control	Purpose
TX CALLSIGN	Your station identifier. Required before transmitting.
TX VOLUME	Controls the transmit-audio level sent to the selected output device.
SEND MESSAGE	Queues the typed message or prepared file for transmission.
STOP (TX)	Stops/cancels transmit work.
IMPORT FILE	Loads a text file into the message box or prepares another file/image for transfer.
ERASE TEXT (TX)	Clears the transmit message box and any prepared file.
RX CALLSIGN	Destination for outgoing messages; CQ is used for general calling/reception.
RX VOLUME	Controls the receive-audio level used by LawnGnomeDX.
Waterfall	Shows the incoming audio spectrum and is useful for confirming that radio audio is present.

6. Send and Receive Files or Images

Sending

16. Press IMPORT FILE and choose the file.
17. Recognized text files are loaded into the TX text box and are sent as normal text.
18. Other files are prepared as a binary transfer. LawnGnomeDX shows a preparation summary in the TX pane.
19. Press SEND MESSAGE to queue the prepared file.
 - The program supports original binary files up to 8 MiB, but a file can still be rejected if it cannot be divided into valid transmissions within the selected time limit.
 - Images may be reduced automatically so they are practical to transmit.
 - After a binary file is prepared, press SEND MESSAGE without editing the preparation text. ERASE TEXT cancels the prepared file.

Receiving

- A received file appears in the RX pane as a clickable filename.
- Click the filename to choose where to save the file. Received files are not automatically saved.
- If image display is enabled, supported images can be previewed directly in the RX pane.

7. Transmission Settings

Open Settings > Transmission Settings.

Setting	Meaning
MAX TRANSMIT TIME	60, 90, or 120 seconds. This controls the per-transmission time target.
COOLDOWN DURATION	A multiple of the preceding transmit time. LawnGnomeDX waits through the cooldown before another transmission.
DATA MODE	BPSK or QPSK for the fast binary/file modem. BPSK is the default.
SPEED	Manual symbol timing when Auto Speed is turned off.
AUTO SPEED	Lets LawnGnomeDX choose a suitable speed automatically. It is enabled by default.

Good starting point: Leave AUTO SPEED enabled unless you have a specific reason to set the speed manually.

8. Receive Settings

Open Settings > Receive Settings.

- Show Images in RX Pane — displays received image previews when possible.
- ENABLE RX AUDIO AGC / CLIP GUARD — automatically manages receive level inside LawnGnomeDX and protects downstream decoding from excessive level.
- RX ECHO MODE — places completed local transmissions into the RX pane as a local record.

Clipping: If the radio, interface, or operating system is already clipping before the audio reaches LawnGnomeDX, reduce that external input level. Software clip protection cannot recover detail that was already clipped upstream.

9. Sound Devices and Language

Sound Devices

- Open Sound > Sound Devices.
- INPUT (RX) is the sound device carrying audio from the radio into LawnGnomeDX.
- OUTPUT (TX) is the sound device carrying LawnGnomeDX transmit audio toward the radio.
- Apply changes the devices for the current session. Save also remembers them for the next launch.
- The modem audio format is fixed at 48 kHz, mono, 16-bit audio.

If a selected RX device is disconnected, LawnGnomeDX keeps trying to reconnect to it. It does not silently switch to another input device. You can choose a different device from the Sound menu if needed.

Alert Sound

The Sound Devices window also provides optional receive-alert controls. If you use alerts, choose an alert output that is different from the modem TX output so the alert cannot be mixed into transmitted modem audio.

Language

Open Settings > Language Selection to change the interface language. The choice is saved for later launches.

10. Optional Encryption

Encryption is disabled by default. Open Settings > Advanced Settings > Encryption if you intend to use it.

20. Enter a Privacy Key, or use GEN KEY to create one.
21. The other station must use the same Privacy Key to decode the encrypted transmission.
22. Enable encryption and read the legal notice shown by LawnGnomeDX. The program requires acceptance before encryption can be enabled.

Legal caution: Use encryption only where it is lawful for your radio service and location. LawnGnomeDX specifically warns that, in the U.S. Amateur Radio Service, transmitting messages encoded to obscure their meaning may violate 47 CFR Part 97 except for limited exceptions. The program's notice is not legal advice.

11. Troubleshooting

Problem	Check this first
Nothing is received	Confirm INPUT (RX), radio audio routing, RX volume, and that the waterfall shows incoming audio. For CQ traffic, use CQ in RX CALLSIGN. For direct traffic, make sure your own callsign is entered in TX CALLSIGN.
Transmit stays queued	The channel may be busy or a cooldown may still be active. LawnGnomeDX waits before transmitting.
Transmission will not start	Confirm OUTPUT (TX) and also confirm the RX input is available. LawnGnomeDX needs receive audio to check the channel before transmitting.
Audio is clipping	Reduce the radio/interface/OS input level. You can also enable RX AUDIO AGC / CLIP GUARD.

File will not send	Try a smaller file or image, or increase MAX TRANSMIT TIME. A file must fit into valid transmission chunks.
Encrypted message will not decode	Confirm both stations are using the exact same Privacy Key. A damaged encrypted transfer is rejected rather than displayed as corrupted text.
Live TX/RX unavailable when running source	Make sure PyAudio/PortAudio, Tkinter, and NumPy are installed for the Python environment.

12. Recommended Starting Setup

- Select the correct RX input and TX output sound devices.
- TX CALLSIGN: your station callsign.
- RX CALLSIGN: CQ for general calling/reception, or the station you want to contact.
- AUTO SPEED: On.
- MAX TRANSMIT TIME: 90 seconds.
- DATA MODE: BPSK.
- RX AUDIO AGC / CLIP GUARD: On.
- Encryption: Off unless you intentionally need it and may lawfully use it.

Normal operation: Once audio routing and callsigns are correct, everyday use is simply: type a message, press SEND MESSAGE, and leave LawnGnomeDX running to receive replies.